

Track 2 Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `Track 2H` mixture-density heads campaign into the official `Track 2` offline verification matrix.

Decision

- `Track 2H` mixture-density heads are verified as an exploratory dispersion-aware baseline.
- No MDN candidate is promoted over the current accepted `Track 2` leaders.
- The strongest MDN forward candidate is `track2h_mdn_k3_Fw`.
- The strongest MDN backward candidate is `track2h_mdn_k2_Bw`.
- The strongest MDN global candidate is `track2h_mdn_k2_global`.
- The best MDN backward candidate improves over the previous quantile/probabilistic and robust-loss `Track 2H` backward branches, but it remains behind the accepted periodic temporal backward leader.
- The project continues to maintain direction-parallel `Fw`, `Bw`, and `global` surfaces instead of collapsing them into one scalar ranking.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary:
`output/validation_checks/track2_reference_comparison/2026-06-13-17-24-53__track2_full_directional_family_matrix_track2h_mixture_density_heads_track2_refresh_2026_06_13/validation_summary.yaml` ;
- per-condition metrics:
`output/validation_checks/track2_reference_comparison/2026-06-13-17-24-53__track2_full_directional_family_matrix_track2h_mixture_density_heads_track2_refresh_2026_06_13/per_condition_metrics.csv` ;
- best-model collage report:
`doc/reports/analysis/track2/best_model_collage_report/[2026-06-13]/track2_best_model_collage_report.md` ;
- multi-model curve comparison report:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-13]/track2_multi_model_curve_comparison_report.md` ;
- operator launch logs:
`output/validation_checks/track2_operator_launch_logs/2026-06-13-17-24-48_track2h_mixture_density_heads_track2_refresh_2026_06_13/` .

Candidate Refresh

The refresh added 6 registry-backed Track 2H MDN candidates covering two- and three-component mixture heads across the `global` , `Fw` , and `Bw` surfaces.

Surface	Profile	Deterministic Curve	Candidate
<code>global</code>	<code>mdn_k2</code>	mixture expectation	<code>track2h_mdn_k2_global</code>
<code>Fw</code>	<code>mdn_k2</code>	mixture expectation	<code>track2h_mdn_k2_Fw</code>
<code>Bw</code>	<code>mdn_k2</code>	mixture expectation	<code>track2h_mdn_k2_Bw</code>
<code>global</code>	<code>mdn_k3</code>	mixture expectation	<code>track2h_mdn_k3_global</code>
<code>Fw</code>	<code>mdn_k3</code>	mixture expectation	<code>track2h_mdn_k3_Fw</code>
<code>Bw</code>	<code>mdn_k3</code>	mixture expectation	<code>track2h_mdn_k3_Bw</code>

The matrix source group is `track2h_mixture_density_heads_registry` . The matrix now contains 153 candidates.

Verification Rules

Surface	Training or Archive Scope	Evaluation Scope
<code>Fw</code>	forward-only training or archive	forward Track 2 curves only

Surface	Training or Archive Scope	Evaluation Scope
Bw	backward-only training or archive	backward Track 2 curves only
global	forward and backward training together	both directions, reported by direction and combined

The MDN heads are not compared as raw mixture tensors. The official matrix evaluates only the deterministic TE curve exposed by each checkpoint: the mixture expectation computed from component logits and component means.

Current Leaders

Overall Direction Leaders

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912
backward	periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
global forward	periodic_gru_sequence_global	0.002777	0.003025	6.267	13.580
global backward	periodic_gru_sequence_global	0.002630	0.002872	6.010	12.693

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	tree_Bw	0.003258	0.003651	7.051	14.116
global forward	tree_global	0.002998	0.003364	6.590	11.601
global backward	tree_global	0.003290	0.003702	7.118	13.703

Track 2H Mixture Density Result

Surface	Candidate	Profile	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Bw	track2h_mdn_k2_Bw	mdn_k2	0.002668	0.002947	5.880	15.593
Bw	track2h_mdn_k3_Bw	mdn_k3	0.002730	0.003009	6.049	15.512
global forward	track2h_mdn_k2_global	mdn_k2	0.003263	0.003582	7.261	14.354
Fw	track2h_mdn_k3_Fw	mdn_k3	0.003226	0.003487	7.164	11.702
Fw	track2h_mdn_k2_Fw	mdn_k2	0.003329	0.003593	7.388	12.771
global forward	track2h_mdn_k3_global	mdn_k3	0.003415	0.003713	7.594	15.012
global backward	track2h_mdn_k3_global	mdn_k3	0.003701	0.004023	8.129	16.261

Surface	Candidate	Profile	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
global backward	track2h_mdn_ k2_global	mdn_k2	0.003735	0.004073	8.194	15.436

The strongest MDN `Bw` candidate improves over the best quantile/probabilistic `Bw` candidate from `0.002935` to `0.002668` Track 2 MAE. It also improves over the best robust-loss `Bw` candidate from `0.003078` to `0.002668` Track 2 MAE. This makes `track2h_mdn_k2_Bw` the strongest Track 2H backward surface so far.

The same improvement does not carry over to `Fw` or `global`. The strongest MDN `Fw` candidate remains behind the best probabilistic `Fw` candidate, and both MDN global candidates remain weaker than the best probabilistic global surface. The campaign-level mixture diagnostics also showed effective component counts near `1.0`, so this official refresh does not prove a stable learned multimodal TE distribution.

Visual Evidence

The `2026-06-13` collage and overlay bundles were regenerated after the MDN refresh. They include explicit Track 2H mixture-density sections for `Fw`, `Bw`, and `global` collage evidence, plus explicit `Fw` and `Bw` MDN overlay sections.

The visual source-coverage validator passed with `track2h_mixture_density_heads_registry` exposed in both visual reports: `collage=6`, `overlay_forward=2`, and `overlay_backward=2`.

The visual package supports the matrix decision: MDN is a useful dispersion-aware backward branch, but it does not replace the strongest periodic temporal branch or the forward paper-reference leader.

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-13	Track 2H mixture-density heads refresh	6 MDN <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 153 - candidate matrix	dated collage and overlay bundles regenerated with MDN sections	verified exploratory baseline; not promoted
2026-06-12	Track 2H quantile/probabilistic refresh	6 quantile and Gaussian <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 147 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted
2026-06-11	Track 2H robust-loss dispersion-aware refresh	9 robust <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 141 - candidate matrix	dated collage and overlay bundles regenerated with Track 2H sections	verified exploratory baseline; not promoted
2026-06-10	Track 2G curve-aware training refresh	12 pointwise-control, centered-shape, offset, and full-composite <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 132 - candidate matrix	dated collage and overlay bundles regenerated with Track 2G sections	verified exploratory baseline; not promoted
2026-06-08	Track 2F-bis harmonic-offset probe refresh	6 clean and harmonic <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates plus 3 rechecked Track 2F candidates	included in the 120 - candidate matrix	dated collage and overlay bundles regenerated with Track 2F-bis sections	verified exploratory baseline; not promoted
2026-06-04	Track 2F offset-aware probe refresh	3 <code>global</code> , <code>Fw</code> , and <code>Bw</code> sequential residual offset candidates	included in the 114 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted
2026-05-28	Wave 2C residual harmonic temporal hybrid refresh	18 <code>global</code> , <code>Fw</code> , and <code>Bw</code> residual harmonic	included in the 111 - candidate matrix	collage and overlay reports refreshed with Wave 2C sections	verified exploratory baseline; not promoted over Wave 2B or

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
		GRU/LSTM candidates			accepted Track 2 baselines

Closeout Decision

Track 2H mixture-density heads do not change the accepted Track 2 baseline. The current direction-parallel decision remains:

- Fw : rcim_retuned_GBM19_Fw remains the strongest overall forward candidate. Within the MDN package, track2h_mdn_k3_Fw is the strongest forward-only candidate, but it trails the best probabilistic forward branch.
- Bw : periodic_gru_sequence_Bw remains the strongest accepted repository-owned backward candidate. Within the MDN package, track2h_mdn_k2_Bw is the strongest backward candidate and the strongest Track 2H backward result so far.
- global : periodic_gru_sequence_global remains the strongest accepted bidirectional neural candidate. Within the MDN package, track2h_mdn_k2_global is the strongest global surface, but it trails the best probabilistic global candidate. The next modeling step should not be another MDN-only variant. The evidence supports moving to the first real Wave 3 hybrid structured campaign, while keeping robust and probabilistic losses as candidate ingredients for later multi-task / multi-head integration. A latent-state / hysteresis-aware Track 2H branch remains justified if the next decision is to pursue experimental state compensation before structured harmonic priors.